PET-VET
NEW FIRE ALARM SYSTEM
PROJECT #60-004247

1001 SOUTH DOUGLAS BOULEVARD
MIDWEST CITY, OKLAHOMA 73130

OCCUPANCY:	B (OFFICE)
CONSTRUCTION:	TYPE IIB
DESCRIPTION:	EXISTING BUILDING REMODEL
AREA:	22,791 S.F.
FIRE SPRINKLERS:	FULLY SPRINKLED
AHJ:	MIDWEST CITY FIRE MARSHAL
CODES:	2018 IRC (INTERNATIONAL RESIDENTIAL CODE WITH OUBCC AMENDMENTS) 2018 IBC (INTERNATIONAL BUILDING CODE WITH OUBCC AMENDMENTS) 2018 IEC (INTERNATIONAL ENERGY CODE WITH OUBCC AMENDMENTS) 2018 IPC (INTERNATIONAL PLUMBING CODE) 2018 IMC (INTERNATIONAL MECHANICAL CODE) 2018 IFGC (INTERNATIONAL FUEL GAS CODE 2020 NEC, NFPA 70(WITH OUBCC AMENDMENTS, MAY 1, 2023)

FIRETROL
Protection Systems
fire protection and life safety specialists
108 NW 132ND STREET
OKLAHOMA CITY, OKLAHOMA 73114
CALL (405) 752-2330 / FAX (405) 752-2297
WWW.FIRETROL.NET
OK LICENSE #863

SCOPE OF WORK:
NEW FIRE ALARM SYSTEM FOR THE FOLLOWING LOCATION:
PET-VET
1001 SOUTH DOUGLAS BOULEVARD, MIDWEST CITY, OKLAHOMA 73130

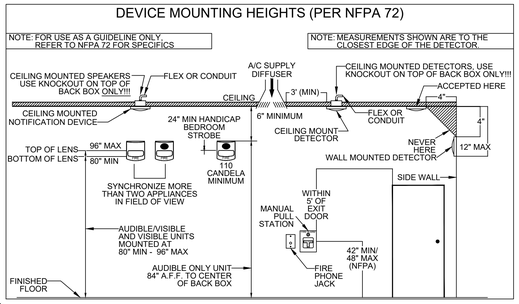
NEW FIRE ALARM SYSTEM SHALL CONSIST OF THE FOLLOWING:
FIRE ALARM CONTROL PANEL WITH BATTERY BACK-UP
REMOTE LCD ANNUNCIATOR
REMOTE POWER SUPPLY WITH BATTERY BACK-UP
CELLULAR COMMUNICATOR
FIRE ALARM DOCUMENT BOX
SURGE SUPPRESSOR, 120 VAC

MANUAL PULL STATION
PHOTOELECTRIC SMOKE DETECTORS
MONITOR MODULES

CEILING MOUNTED HORN/STROBES
CEILING MOUNTED STROBES
WALL MOUNTED HORN STROBE, WEATHERPROOF

General Notes
1. 120 VAC input power for the Fire Alarm Equipment shall be dedicated circuits with a locking breaker properly labeled.
2. 120 VAC is provided by others. Coordinate 120 VAC with Electrical Contractor.
3. 120 VAC is not permitted in the same conduit with low voltage wiring.
4. Field verify locations of 120 VAC and circuit numbers.
5. AHJ may require a separate permit for all 120 VAC work performed on project.
6. Wiring methods and materials shall conform with all applicable sections of NFPA & NEC codes.
7. All pull boxes shall be sized & labeled in accordance to NEC.
8. Installation shall conform to all requirements of applicable Fire Alarm & Electrical codes.
9. All installation will be done by licensed technicians meeting all qualifications of the State of Oklahoma and/or the Authority having Jurisdiction.
10. All wiring, including shielding must be dry and free of shorts and ground faults.
11. Conduit for fire alarm system shall be sized to not exceed 30% fill capacity.
12. All devices in the alarm system shall be compatible and installed per manufacturer's specifications.
13. All strobes shall be synchronized.
14. Wall mount horn/strobes and strobes shall be mounted with the center of the back back boxes at 82" above finished floor.
15. No smoke detector head shall be installed prior to the area being finished & cleaned.
16. Manual fire alarm pull stations shall be mounted at 45" +/- 3" above the finished floor to the operable part of the device & shall be located within 5' of the exit doorway opening.
17. 24 Hour Central Station Monitoring shall be provided by the owner.
18. Local fire authority / Authority Having Jurisdiction shall witness final test.
19. Ceiling mounted devices will require that the back box knock out used is on the top of the back box only.

DEVICE MOUNTING HEIGHTS (PER NFPA 72)



FIRE ALARM SYMBOLS

SYMBOL	SYMBOL DESCRIPTION	MANUFACTURER	MODEL NUMBER(S)	MOUNTING	QTY
FACP	FIRE ALARM CONTROL PANEL	FIREWARDEN	NFW-100X	RED ENCLOSURE	001
	BATTERIES- 7.5 AH	INTERSTATE	SLA1080	MOUNTS IN ENCLOSURE	002
LCD	REMOTE ANNUNCIATOR LCD	NOTIFIER	N-ANN-80	3 GANG ELECTRICAL BOX	001
RPS	REMOTE POWER SUPPLY 10 AMP	NOTIFIER	HPF-PS10	SURFACE MOUNT	001
	BATTERY-7.5 AH	INTERSTATE	SLA1080	MOUNTS IN HPF-PS10B	002
CELL	FIRE ALARM COMMUNICATOR	HONEYWELL	HW-AV-ENC	SURFACE MOUNT	001
	CELLULAR SOLE PATH	HONEYWELL	HW-AV-LTE-M-2	MOUNTS IN HW-AV-ENC	001
DOC	FIRE ALARM DOCUMENT BOX -	SPACE AGE	SSU00685	SURFACE MOUNT	001
SS	TRANSIENT VOLTAGE SURGE SUPPRESSOR RATED FOR 120 VOLTS	DITEK	DTK-120SLR	SURFACE MOUNT	001
F	MANUAL PULL STATION DOUBLE ACTION	FIREWARDEN	NOT-BG12LX	SINGLE GANG	001
P	PHOTOELECTRIC SMOKE DETECTOR STANDARD BASE	FIREWARDEN	NP-200 (SMOKE) B300-6 (BASE)	ELECT BOX >1.5" DEEP	002
M	INTELLIGENT MODULE MONITOR SINGLE CIRCUIT	FIREWARDEN	NMM-100	4" SQ BOX, 1.25" DEEP	005
15CD	FIRE ALARM LED HORN/STROBE CEILING MOUNT, WHITE, "FIRE"	SYSTEM SENSOR	PC2WLED	4" SQ BOX, 1.50" DEEP	042
15CD	FIRE ALARM LED STROBE CEILING MOUNT, WHITE, "FIRE"	SYSTEM SENSOR	SCWLED	4" SQ BOX, 1.50" DEEP	032
15CD FDCW	FIRE ALARM HORN/STROBE, WEATHER PROOF WALL MOUNT, RED, CLEAR LENS, "FIRE"	SYSTEM SENSOR	P2GRKLED	COMES WITH BACK BOX	001
FS	FIRE SPRINKLER SYSTEM FLOW SWITCH	BY OTHERS	BY OTHERS	N/A	001
TS	FIRE SPRINKLER SYSTEM TAMPER SWITCH	BY OTHERS	BY OTHERS	N/A	001

FIRE ALARM WIRE (STANDARD)

WIRE APPEARANCE	SYMBOL DESCRIPTION	WIRE DESCRIPTION
120 VAC	120 VAC FOR CONTROL PANELS PROVIDED BY OTHERS	PROVIDED BY OTHERS #12 SOLID THHN
P — P	24 VDC FOR AUXILIARY DEVICES SUPPLIED BY FIRE ALARM CONTRACTOR	#14/2 AWG SOLID COPPER, NON-SHIELDED FPLP BLACK JACKET, RED (+) AND BLACK (-)
DAL — DAL	FIRE ALARM DAL SUPPLIED BY FIRE ALARM CONTRACTOR	#14/2 AWG SOLID COPPER, NON-SHIELDED FPLP GREEN JACKET, RED (+) AND BLACK (-)
X — X	FIRE ALARM ANNUNCIATOR SUPPLIED BY FIRE ALARM CONTRACTOR	#16/4 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/WHITE STRIPE, RED (+) AND BLACK (-)
L — L	SIGNALING LINE CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/YELLOW STRIPE, RED (+) AND BLACK (-)
D — D	INITIATING DEVICE CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/YELLOW STRIPE, RED (+) AND BLACK (-)
RT — RT	REMOTE TEST STATION SUPPLIED BY FIRE ALARM CONTRACTOR	#16/4 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/WHITE STRIPE, RED (+) AND BLACK (-)
V — V	VISUAL NOTIFICATION CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#14/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/BLUE STRIPE, RED (+) AND BLACK (-)
A — A	SPEAKER CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP ORANGE JACKET, RED (+) AND BLACK (-)

SEQUENCE OF OPERATIONS MATRIX

		ACTUATE AUDIBLE & VISUAL SIGNAL INDICATOR FOR ALARM	TRANSMIT ALARM CONDITION TO SUPERVISING STATION	ACTUATE AUDIBLE & VISUAL SIGNAL INDICATOR FOR SUPERVISORY	TRANSMIT SUPERVISORY CONDITION TO SUPERVISING STATION	ACTUATE AUDIBLE & VISUAL SIGNAL INDICATOR FOR TROUBLE	TRANSMIT TROUBLE CONDITION TO SUPERVISING STATION	DISPLAY CHANGE FOR ALL FIRE ALARM CONTROL EQUIPMENT	TRANSMIT POINT IDENTIFICATION TO SUPERVISING STATION	ACTUATE NOTIFICATION APPLIANCES FOR ENTIRE BUILDING (TEMPORAL 3)	ACTUATE FDC HORN/STROBE
01	MANUAL PULL STATION	X	X					X	X	X	
02	SMOKE DETECTOR	X	X					X	X	X	
03	DUCT DETECTOR AT AHU			X	X			X	X		
04	FLOW SWITCH	X	X					X	X	X	X
05	TAMPER SWITCH			X	X			X	X		
06	PANEL LOW BATTERY CONDITION					X	X	X	X		
07	AC POWER FAILURE					X	X	X	X		
08	GROUND FAULT					X	X	X	X		
09	SHORT CIRCUIT					X	X	X	X		
10	OPEN CIRCUIT					X	X	X	X		
11	ANY SYSTEM TROUBLE					X	X	X	X		

FIRE ALARM DEVICE DESIGNATIONS

FIELD DEVICE ADDRESSING:
- F — L1Mxxx ← MODULE ADDRESS
- P — L1Dxxx ← DETECTOR ADDRESS
- 95CD ← CANDELA RATING
- H — V1x-x-xx ← STROBE CIRCUIT DESIGNATION

SLC DEVICE ADDRESSING:
- L1M or Dxxx
  - DEVICE #
  - LOOP #

STROBE ADDRESSING:
- V1x-xx-xx
  - DEVICE #
  - CIRCUIT #
  - REMOTE POWER SUPPLY NUMBER

Fire Alarm Drawing Index

FA-0.00	COVER
FA-2.01	FLOOR PLAN & DEVICE LAYOUT
FA-3.01	CALCULATIONS AND WIRING DETAILS

REVISIONS
A	B	C
A	B	C

REVIEWED BY:
BRIAN T. LATTEA - NICET #105235
FIRE ALARM SYSTEMS - LEVEL III

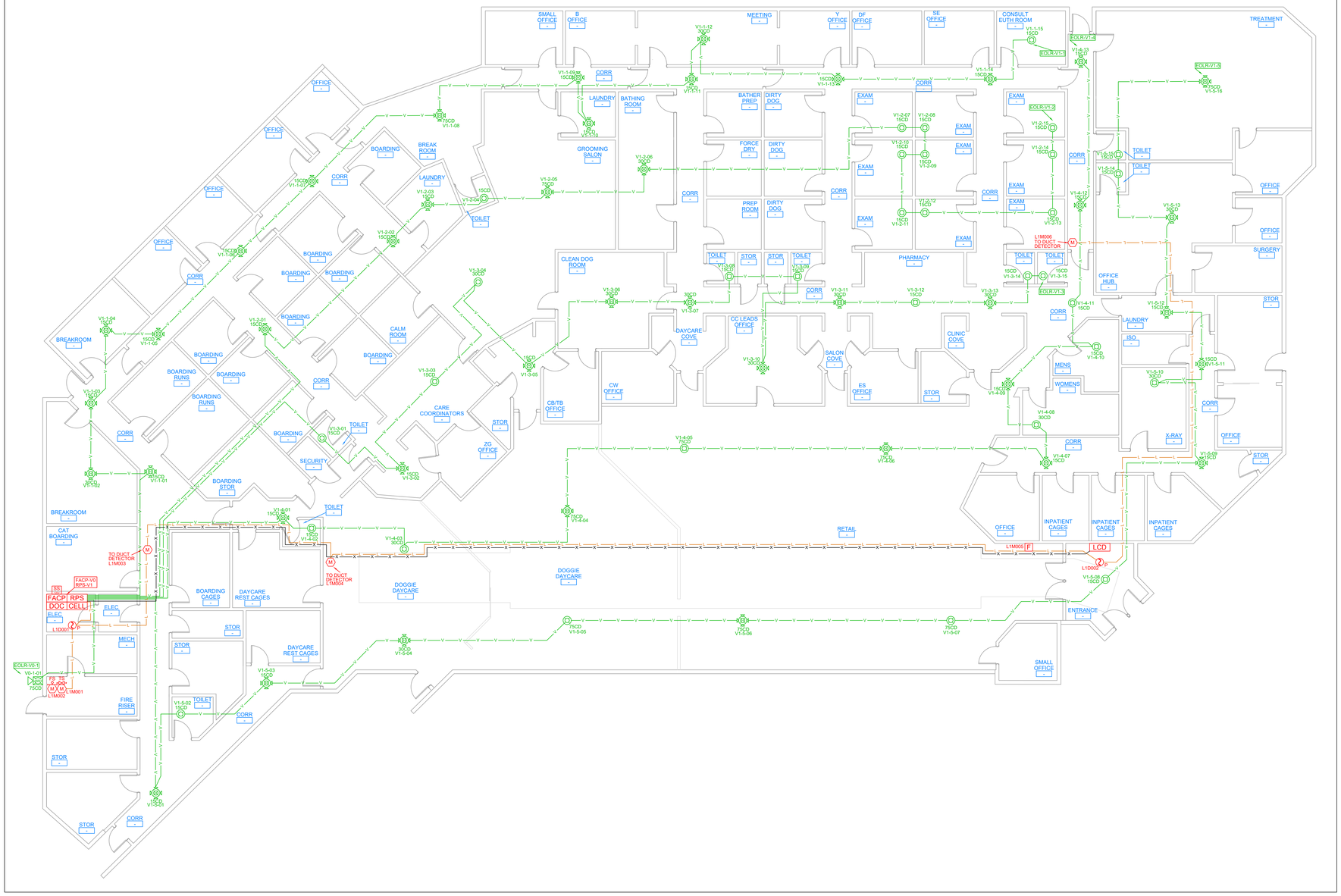
DATE:	2026-07-09
PROJECT NO.:	60-004247
DRAWN BY:	R.TOFFOLI
CHECKED BY:	B.T.LATTEA
SPEC SECTION:	-------
ISSUE DATE:	-------

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FIRE ALARM SYSTEM

COVER

FA
0.00



FIRE ALARM SYMBOLS

SYMBOL	SYMBOL DESCRIPTION	MANUFACTURER	MODEL NUMBER(S)	MOUNTING	QTY
FACP	FIRE ALARM CONTROL PANEL BATTERIES- 7.5 AH	FIREWARDEN	NFW-100X	RED ENCLOSURE	001
		INTERSTATE	SLA1080	MOUNTS IN ENCLOSURE	002
LCD	REMOTE ANNUNCIATOR LCD	NOTIFIER	N-ANN-80	3 GANG ELECTRICAL BOX	001
RPS	REMOTE POWER SUPPLY 10 AMP BATTERY-7.5 AH	NOTIFIER	HPF-PS10	SURFACE MOUNT	001
		INTERSTATE	SLA1080	MOUNTS IN HPF-PS10B	002
CELL	FIRE ALARM COMMUNICATOR CELLULAR SOLE PATH	HONEYWELL	HW-AV-ENC	SURFACE MOUNT	001
		HONEYWELL	HW-AV-LTE-M-2	MOUNTS IN HW-AV-ENC	001
DOC	FIRE ALARM DOCUMENT BOX -	SPACE AGE	SSU00685	SURFACE MOUNT	001
SS	TRANSIENT VOLTAGE SURGE SUPPRESSOR RATED FOR 120 VOLTS	DITEK	DTK-120SLR	SURFACE MOUNT	001
F	MANUAL PULL STATION DOUBLE ACTION	FIREWARDEN	NOT-BG12LX	SINGLE GANG	001
P	PHOTOELECTRIC SMOKE DETECTOR STANDARD BASE	FIREWARDEN	NP-200 (SMOKE) B300-6 (BASE)	ELECT BOX +1.5" DEEP	002
M	INTELLIGENT MODULE MONITOR SINGLE CIRCUIT	FIREWARDEN	NMM-100	4" SQ BOX, 1.25" DEEP	005
15CD	FIRE ALARM LED HORN/STROBE CEILING MOUNT, WHITE, "FIRE"	SYSTEM SENSOR	PC2WLED	4" SQ BOX, 1.50" DEEP	042
15CD	FIRE ALARM LED STROBE CEILING MOUNT, WHITE, "FIRE"	SYSTEM SENSOR	SCWLED	4" SQ BOX, 1.50" DEEP	032
75CD	FIRE ALARM HORN/STROBE, WEATHER PROOF WALL MOUNT, RED, CLEAR LENS, "FIRE"	SYSTEM SENSOR	P2GRKLED	COMES WITH BACK BOX	001
FS	FIRE SPRINKLER SYSTEM FLOW SWITCH	BY OTHERS	BY OTHERS	N/A	001

FIRE ALARM WIRE (STANDARD)

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—— 120 VAC ——	120 VAC FOR CONTROL PANELS PROVIDED BY OTHERS	PROVIDED BY OTHERS #12 SOLID THHN
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—X—X—	FIRE ALARM ANNUNCIATOR SUPPLIED BY FIRE ALARM CONTRACTOR	#16/4 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/WHITE STRIPE, RED (+) AND BLACK (-)
—L—L—	SIGNALING LINE CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/YELLOW STRIPE, RED (+) AND BLACK (-)
—D—D—	INITIATING DEVICE CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/YELLOW STRIPE, RED (+) AND BLACK (-)
—RT—RT—	REMOTE TEST STATION SUPPLIED BY FIRE ALARM CONTRACTOR	#16/4 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/WHITE STRIPE, RED (+) AND BLACK (-)
—V—V—	VISUAL NOTIFICATION CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#14/2 AWG SOLID COPPER, NON-SHIELDED FPLP RED JACKET/BLUE STRIPE, RED (+) AND BLACK (-)
—A—A—	SPEAKER CIRCUIT SUPPLIED BY FIRE ALARM CONTRACTOR	#16/2 AWG SOLID COPPER, NON-SHIELDED FPLP ORANGE JACKET, RED (+) AND BLACK (-)

FIRE ALARM DEVICE DESIGNATIONS

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95CD ← CANDELA RATING
H V1x-x-xx ← STROBE CIRCUIT DESIGNATION

SLC DEVICE ADDRESSING:

L1M or Dxxx
- DEVICE #
- LOOP #

STROBE ADDRESSING:

V1x-x-xx
- DEVICE #
- CIRCUIT #
- REMOTE POWER SUPPLY NUMBER

N

FLOOR PLAN & DEVICE LAYOUT

1/8"=1'-0"

1' 3' 5' 7' 9' 11' 13' 15' 17' 19'
0 2' 4' 6' 8' 10' 12' 14' 16' 18' 20'

REVISIONS

A	
B	
C	

REVIEWED BY:
BRIAN T. LATTEA - NICET #105235
FIRE ALARM SYSTEMS - LEVEL III

DATE	2026-07-09
PROJECT NO.	60-0042447
DRAWN BY	R.TOFFOLI
CHECKED BY	B.T.LATTEA
SPEC SECTION	------
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PET-VET
NEW FIRE ALARM SYSTEM
1001 SOUTH DOUGLAS BOULEVARD
MIDWEST CITY, OKLAHOMA 73130
FIRE ALARM SYSTEM

FLOOR PLAN & DEVICE LAYOUT

FA
2.01

BATTERY CALCULATIONS FOR FIRE ALARM CONTROL PANEL							
QUANTITY	MODEL #	DESCRIPTION		STANDBY CURRENT	TOTAL STANDBY	ALARM CURRENT	TOTAL ALARM
PANEL							
1	NFW-100X	NOTIFIER FIRE ALARM CONTROL PANEL		0.141000	0.141000	0.141000	0.141000
1	HW-AV-LTE-M-2	CLSS DUAL PATH COMMUNICATOR		0.060000	0.060000	0.200000	0.200000
2	SLA1080	BATTERY, 12 VOLT, 7.5 AMP/HOUR		0.000000	0.000000	0.000000	0.000000
ANNUNCIATORS							
1	N-ANN-80	REMOTE LCD ANNUNCIATOR, 80 CHARACTERS		0.040000	0.040000	0.040000	0.040000
INITIATING DEVICES							
1	NOT-BG12LX	MANUAL STATION, DUAL ACTION		0.000375	0.000375	0.005000	0.005000
2	NP-200	SMOKE DETECTOR, PHOTOELECTRIC		0.000200	0.000400	0.004500	0.009000
2	B300-6	DETECTOR MOUNTING BASE		0.000000	0.000000	0.000000	0.000000
5	NMM-100	MONITOR MODULE		0.000350	0.001750	0.005000	0.025000
NOTIFICATION APPLIANCES							
1	NAC 1	NOTIFICATION APPLIANCE CIRCUIT 1				0.087000	0.087000
1	NAC 2	NOTIFICATION APPLIANCE CIRCUIT 2				0.000000	0.000000
1	NAC 3	NOTIFICATION APPLIANCE CIRCUIT 3				0.000000	0.000000
1	NAC 4	NOTIFICATION APPLIANCE CIRCUIT 4				0.000000	0.000000
1	NAC 5	NOTIFICATION APPLIANCE CIRCUIT 5				0.000000	0.000000
1	NAC 6	NOTIFICATION APPLIANCE CIRCUIT 6				0.000000	0.000000
1	NAC 7	NOTIFICATION APPLIANCE CIRCUIT 7				0.000000	0.000000
		TOTAL STANDBY AND ALARM			0.243525		0.507000

STANDBY TIME:	24	HR	x	0.243525	=	5.844600
ALARM TIME:	15	MIN	x	0.507000	=	0.126750
STANDBY + ALARM TIME:					=	5.971350
BATTERY DISCHARGE FACTOR (20%):					=	1.194270
MINIMUM BATTERY CAPACITY REQUIRED:					=	7.165620
BATTERY CAPACITY PROVIDED:					=	7.500000

VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V0-1									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V0-1-01	P2GRKLED(75)	HRN/STR,WALL,RED,"FIRE",75CD, WP	0.087	33	14	0.087	33	0.014	20.386

BATTERY CALCULATIONS FOR FIRE ALARM POWER SUPPLY #V1							
QUANTITY	MODEL #	DESCRIPTION		STANDBY CURRENT	TOTAL STANDBY	ALARM CURRENT	TOTAL ALARM
PANEL							
1	HPF-PS10B	FIRE ALARM POWER SUPPLY		0.178000	0.178000	0.232000	0.232000
2	SLA1080	BATTERY, 12 VOLT, 7.5 AMP/HOUR		0.000000	0.000000	0.000000	0.000000
NOTIFICATION APPLIANCES							
1	NAC 1	NOTIFICATION APPLIANCE CIRCUIT 1		0.000000	0.000000	0.583000	0.583000
1	NAC 2	NOTIFICATION APPLIANCE CIRCUIT 2		0.000000	0.000000	0.410000	0.410000
1	NAC 3	NOTIFICATION APPLIANCE CIRCUIT 3		0.000000	0.000000	0.408000	0.408000
1	NAC 4	NOTIFICATION APPLIANCE CIRCUIT 4		0.000000	0.000000	0.517000	0.517000
1	NAC 5	NOTIFICATION APPLIANCE CIRCUIT 5		0.000000	0.000000	0.659000	0.659000
1	NAC 6	NOTIFICATION APPLIANCE CIRCUIT 6		0.000000	0.000000	0.000000	0.000000
1	NAC 7	NOTIFICATION APPLIANCE CIRCUIT 7		0.000000	0.000000	0.000000	0.000000
		TOTAL STANDBY AND ALARM			0.178000		2.809000

STANDBY TIME:	24	HR	x	0.178000	=	4.272000
ALARM TIME:	15	MIN	x	2.809000	=	0.702250
STANDBY + ALARM TIME:					=	4.974250
BATTERY DISCHARGE FACTOR (20%):					=	0.994850
MINIMUM BATTERY CAPACITY REQUIRED:					=	5.969100
BATTERY CAPACITY PROVIDED:					=	7.500000

VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V1-1									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V1-1-01	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	41	14	0.035	41	0.007	20.393
V1-1-02	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	19	14	0.073	60	0.022	20.378
V1-1-03	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	21	14	0.108	81	0.044	20.356
V1-1-04	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	23	14	0.143	104	0.075	20.325
V1-1-05	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	18	14	0.178	122	0.109	20.291
V1-1-06	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	29	14	0.213	151	0.162	20.238
V1-1-07	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	26	14	0.248	177	0.221	20.179
V1-1-08	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	36	14	0.335	213	0.360	20.040
V1-1-09	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	38	14	0.370	251	0.468	19.932
V1-1-10	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	18	14	0.405	269	0.549	19.851
V1-1-11	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	35	14	0.440	304	0.674	19.726
V1-1-12	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	17	14	0.478	321	0.773	19.627
V1-1-13	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	39	14	0.513	360	0.931	19.469
V1-1-14	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	35	14	0.548	395	1.091	19.309
V1-1-15	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	23	14	0.583	418	1.228	19.172

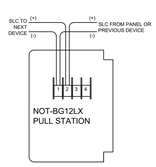
VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V1-2									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V1-2-01	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	81	14	0.035	81	0.014	20.386
V1-2-02	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	40	14	0.070	121	0.043	20.357
V1-2-03	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	18	14	0.105	139	0.074	20.326
V1-2-04	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	19	14	0.123	158	0.098	20.302
V1-2-05	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	31	14	0.210	189	0.200	20.200
V1-2-06	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	29	14	0.248	218	0.272	20.128
V1-2-07	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	60	14	0.266	278	0.373	20.027
V1-2-08	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	13	14	0.284	291	0.417	19.983
V1-2-09	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	14	14	0.302	305	0.464	19.936
V1-2-10	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	13	14	0.320	318	0.513	19.887
V1-2-11	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	19	14	0.338	337	0.574	19.826
V1-2-12	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	13	14	0.356	350	0.628	19.772
V1-2-13	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	31	14	0.374	381	0.718	19.682
V1-2-14	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	19	14	0.392	400	0.790	19.610
V1-2-15	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	14	14	0.410	414	0.855	19.545

VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V1-3									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V1-3-01	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	74	14	0.018	74	0.007	20.393
V1-3-02	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	27	14	0.053	101	0.027	20.373
V1-3-03	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	40	14	0.071	141	0.050	20.350
V1-3-04	SCWLED(30)	LED STROBE, CEILING, WHITE, "FIRE", 30CD	0.022	34	14	0.093	175	0.082	20.318
V1-3-05	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	30	14	0.128	205	0.132	20.268
V1-3-06	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	31	14	0.166	236	0.197	20.203
V1-3-07	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	22	14	0.204	258	0.265	20.135
V1-3-08	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	20	14	0.222	278	0.311	20.089
V1-3-09	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	21	14	0.240	299	0.362	20.038
V1-3-10	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	31	14	0.278	330	0.462	19.938
V1-3-11	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	33	14	0.316	363	0.578	19.822
V1-3-12	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	22	14	0.334	385	0.648	19.752
V1-3-13	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	22	14	0.372	407	0.763	19.637
V1-3-14	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	20	14	0.390	427	0.839	19.561
V1-3-15	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	12	14	0.408	439	0.903	19.497

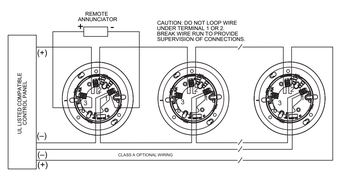
VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V1-4									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V1-4-01	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	57	14	0.035	57	0.010	20.390
V1-4-02	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	15	14	0.053	72	0.019	20.381
V1-4-03	SCWLED(30)	LED STROBE, CEILING, WHITE, "FIRE", 30CD	0.022	29	14	0.075	101	0.038	20.362
V1-4-04	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	44	14	0.162	145	0.118	20.282
V1-4-05	SCWLED(75)	LED STROBE, CEILING, WHITE, "FIRE", 75CD	0.070	40	14	0.232	185	0.216	20.184
V1-4-06	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	44	14	0.319	229	0.368	20.032
V1-4-07	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	38	14	0.354	267	0.476	19.924
V1-4-08	SCWLED(30)	LED STROBE, CEILING, WHITE, "FIRE", 30CD	0.022	16	14	0.376	283	0.536	19.864
V1-4-09	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	21	14	0.411	304	0.630	19.770
V1-4-10	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	27	14	0.429	331	0.716	19.684
V1-4-11	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	21	14	0.447	352	0.793	19.607
V1-4-12	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	26	14	0.482	378	0.918	19.482
V1-4-13	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	35	14	0.517	413	1.076	19.324

VOLTAGE DROP CALCULATIONS FOR FIRE ALARM POWER SUPPLY - CIRCUIT #V1-5									
APPLIANCE NUMBER	APPLIANCE MODEL	APPLIANCE DESCRIPTION	CURRENT DRAW	WIRE DISTANCE	WIRE GA (AWG)	TOTAL CURRENT (A)	TOTAL DISTANCE (FT)	VOLTAGE DROP (V)	VOLTAGE @ DEVICE (V)
V1-5-01	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	55	14	0.035	55	0.010	20.390
V1-5-02	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	26	14	0.053	81	0.022	20.378
V1-5-03	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	26	14	0.088	107	0.047	20.353
V1-5-04	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	35	14	0.126	142	0.090	20.310
V1-5-05	SCWLED(75)	LED STROBE, CEILING, WHITE, "FIRE", 75CD	0.070	38	14	0.196	180	0.178	20.222
V1-5-06	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	39	14	0.283	219	0.312	20.088
V1-5-07	SCWLED(75)	LED STROBE, CEILING, WHITE, "FIRE", 75CD	0.070	45	14	0.353	264	0.470	19.930
V1-5-08	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	39	14	0.371	303	0.567	19.833
V1-5-09	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	43	14	0.406	346	0.708	19.692
V1-5-10	SCWLED(30)	LED STROBE, CEILING, WHITE, "FIRE", 30CD	0.022	31	14	0.428	377	0.813	19.587
V1-5-11	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	20	14	0.463	397	0.926	19.474
V1-5-12	PC2WLED(15)	LED HRN/STR, CEILING, WHITE, "FIRE", 15CD	0.035	23	14	0.498	420	1.054	19.346
V1-5-13	PC2WLED(30)	LED HRN/STR, CEILING, WHITE, "FIRE", 30CD	0.038	36	14	0.536	456	1.232	19.168
V1-5-14	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	26	14	0.554	482	1.346	19.054
V1-5-15	SCWLED(15)	LED STROBE, CEILING, WHITE, "FIRE", 15CD	0.018	13	14	0.572	495	1.427	18.973
V1-5-16	PC2WLED(75)	LED HRN/STR, CEILING, WHITE, "FIRE", 75CD	0.087	36	14	0.659	531	1.764	18.636

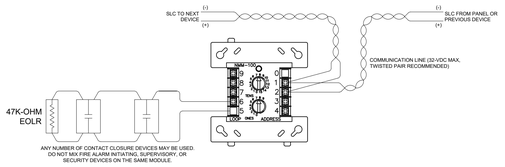
CALCULATIONS



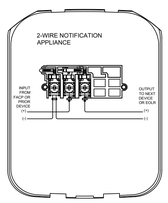
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B300-6 DETECTOR BASE



NMM-100 ADDRESSABLE SINGLE CIRCUIT INPUT MODULE



SYSTEM SENSOR
2-WIRE NOTIFICATION APPLIANCE

WIRING DETAILS

REVISIONS

A			
B			
C			

REVIEWED BY:
BRIAN T. LATTEA - NICET #105235
FIRE ALARM SYSTEMS - LEVEL III

DATE: 2026-07-09
PROJECT NO: 60-0042447
DRAWN BY: R.TOFFOLI
CHECKED BY: B.T.LATTEA
SPEC SECTION: -------
ISSUE DATE: -------

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